

6(a)	$I = I_1 + I_2 + I_3$	B1
	$(V/R) = (V/R_1) + (V/R_2) + (V/R_3)$ or $(I/V) = (I_1/V) + (I_2/V) + (I_3/V)$	B1
	and $1/R = 1/R_1 + 1/R_2 + 1/R_3$	
6(b)(i)	current = $0.49 + 0.45$ $= 0.94 \text{ A}$	A1
6(b)(ii)	$8.0 = (0.94 \times r) + (0.45 \times 16)$	C1
	$r = 0.85 \Omega$	A1
6(c)	$I = Anvq$ $v = (0.45/0.49) \times 2.1 \times 10^{-4}$	C1
	$= 1.9 \times 10^{-4} \text{ m s}^{-1}$	A1
6(d)	total/combined resistance decreases	B1
	(current in battery increases so terminal) potential difference decreases	B1